D.P.M.S.
Artificial Intelligence

Evaluating Supevised Contrastive Learning for noise-robust Speech Emotion Recognition

Course: Deep Learning

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Declaration on the use of Artificial Intelligence

I hereby declare that generative Artificial Intelligence (AI) tools, specifically Large Language Models (Gemini 3.5 Flash - Standard), were utilized during the preparation of this report. The nature and scope of this assistance are detailed below:

- **Mathematical to Code Translation:** AI tools were used as aid to cross-reference theoretical foundations from the source literature against structural `pytorch` classes and operational components.
- **Text Modification and Polishing:** AI assistance was used to refine expressions, correct syntax errors, and smooth technical transitions. The core ideas, literature selection, and methodological descriptions remain entirely my own.

All primary research data, custom convolutional neural network configurations, dataset splitting protocols, and final analytical conclusions were generated entirely through my own engineering work. No section of this report was completely synthesized or fabricated by automated generation platforms.

Abstract

Speech Emotion Recognition (SER) on limited datasets such as RAVDESS often leads to model overfitting. This study investigates a Supervised Contrastive Learning (SupCon) pipeline combined with multi-view SpecAugment to extract invariant emotional representations rather than memorizing specific acoustic frequencies. The experimental results demonstrate that the SupCon architecture increases the clean test accuracy from a 53% baseline to 60% compared to a standard Cross-Entropy approach. Additionally, the proposed methodology exhibits improved classification stability under heavy Gaussian noise degradation.

1 Introduction and Motivation

1.1 The Challenge of Speech Emotion Recognition (SER)

Speech Emotion Recognition (SER) is a notoriously challenging domain within audio processing, primarily due to the high inter-class variance and the subjective nature of human emotional expression. This challenge is heavily exacerbated when working with small-scale datasets such as RAVDESS[1] (Ryerson Audio-Visual Database of Emotional Speech and Song), which contains roughly 1,200 training samples. When deploying standard deep learning architectures on such limited data, neural networks exhibit a strong tendency to bypass the intended task of learning generalized emotional representations. Instead, they exploit the path of least resistance by memorizing the unique acoustic signatures (pitch, timbre, and cadence) of the specific actors in the training set. Consequently, while the model may achieve artificially high training accuracy, its performance drastically degrades when evaluated on unseen actors, exposing weak decision boundaries that fail to capture true semantic emotional features.

1.2 The Flaws of Cross-Entropy and the SupCon Hypothesis

Standard Cross-Entropy (CE) loss, despite being the default standard for classification tasks, is fundamentally limited when applied to highly variable, low-resource datasets. The CE objective strictly penalizes point-wise misclassifications, driving the network to construct linear hyperplanes based on any available discriminative feature. In the context of SER, this aggressive optimization forces the model to latch onto non-semantic acoustic artifacts (background noise or a specific actor's baseline pitch) rather than learning generalized emotional structures. This results in weak decision boundaries which are highly susceptible to distribution shifts or environmental noise.

To address this, we hypothesize that Supervised Contrastive Learning (SupCon) can yield significantly more robust representations. Instead of directly predicting a hard class label, SupCon maps input spectrograms into a normalized embedding space. By pulling heavily augmented, multi-view representations (via Time and Frequency masking) of the same emotion closer together while pushing differing emotions apart, the network is forced to ignore augmentation-induced distortions. We hypothesize that this mechanism decouples the core semantic structure of the emotion from brittle, actor-specific frequencies, thereby producing a classifier that maintains high accuracy even under severe acoustic degradation.

2 Data and Preprocessing

2.1 The RAVDESS Dataset

This study utilizes the speech audio-only subset of the RAVDESS dataset, featuring 24 professional actors (12 male, 12 female) delivering lexically matched statements across eight distinct emotions: Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, and Surprised. The dataset comprises 1,440 highly balanced audio files. To rigorously evaluate model generalization and prevent vocal memorization (data leakage), a strict speaker-independent split was enforced. By holding out specific

[1]https://www.kaggle.com/datasets/uwrfkaggler/ravdess-emotional-speech-audio/data

actors entirely for the test set, the data was partitioned to yield an effective training corpus of approximately 1,200 samples for unbiased evaluation.

2.2 Audio to Mel-Spectrogram Conversion

To leverage the spatial pattern-recognition capabilities of Convolutional Neural Networks (CNNs), the 1D audio waveforms were transformed into 2D Mel-Spectrograms. This process involves computing the Short-Time Fourier Transform (STFT) to extract frequency content over time, followed by mapping the power spectra onto the Mel scale to mimic human auditory perception. The resulting 2D matrix (frequency bins $\times$ time frames) is mathematically processed by the network as a 1-channel grayscale image tensor, providing spatially correlated representation of the acoustic energy.

2.3 Data Augmentation (SpecAugment)

We applied the *SpecAugment* technique to the Mel-Spectrograms during training. This artificially expands the dataset and forces the network to learn invariant features. SpecAugment operates directly on the spectrogram by randomly masking out continuous blocks of frequency channels and time steps. For the baseline model, this acts as a powerful regularizer.

More importantly, SpecAugment drives the Multi-View strategy for the SupCon pipeline. The augmentation generates two differently masked versions of the exact same audio file. The contrastive loss function then forces the CNN to map both distorted inputs to the same coordinate space. This mechanism effectively blinds the network to specific acoustic artifacts and extracts only the underlying emotional structure.

3 Methodology and Architectures

3.1 The Custom CNN Backbone

To extract spatial features from the Mel-Spectrograms, a custom, lightweight 2D Convolutional Neural Network (CNN) was designed. The architecture consists of four sequential convolutional blocks, each utilizing 3×3 kernels, Batch Normalization, Leaky ReLU activation, and 2×2 Max Pooling. The network was intentionally kept shallow. Deploying massively parameterized models on a dataset of $\sim$1,200 samples drastically increases the risk of overfitting.

3.2 Model A: The Cross-Entropy Baseline

To establish a rigorous benchmark, Model A was trained end-to-end using standard Cross-Entropy loss. The 4-layer CNN backbone was flattened and connected to a shallow, two-layer dense classification head. The model was optimized using the Adam optimizer to directly map the input spectrograms to one of the eight discrete emotion classes. This baseline serves to quantify the upper limit of traditional, label-driven classification on the RAVDESS dataset.

3.3 Model B: The SupCon Pipeline

Stage 1: Contrastive Clustering

The core methodology relies on the two-phase training of Model B. In Stage 1, the CNN backbone connects to a non-linear projection head. Each batch of spectrograms undergoes a Multi-View strategy. The SpecAugment module processes the data twice to generate two heavily masked variations of the same audio. The Supervised Contrastive Loss function is then applied to these projected embeddings. This forces the network to pull different views of the same emotion into tightly packed clusters. Simultaneously, it repels the embeddings of different classes. Crucially, this stage focuses entirely on representation learning. The model structures the feature space without outputting explicit class predictions.

Stage 2 & 3: Linear Probing and Fine-Tuning

Once the clusters are firmly established, the projection head is discarded. In Stage 2 (Linear Probing), the CNN encoder is strictly frozen, and a single linear classification layer is attached and trained. This proves the linear separability of the contrastively learned clusters. Finally, in Stage 3 (End-to-End Fine-Tuning), the entire network is unfrozen. Differential learning rates are applied during this phase. A highly conservative learning rate is assigned to the encoder to protect the established clusters. Meanwhile, the classifier head utilizes a standard learning rate. This final micro-adjustment binds the feature extractor and the classifier, yielding the ultimate predictive model.

4 Experiments and Results

4.1 Standard Evaluation (Clean Accuracy)

Emotion	Precision	Recall	F1-Score	Support
Neutral	0.43	0.38	0.40	16
Calm	0.50	0.69	0.58	32
Happy	0.23	0.16	0.19	32
Sad	0.25	0.22	0.23	32
Angry	0.73	0.84	0.78	32
Fearful	0.61	0.34	0.44	32
Disgust	0.56	0.75	0.64	32
Surprised	0.76	0.81	0.79	32
Accuracy			**0.53**	**240**
Macro Avg	0.51	0.52	0.51	240
Weighted Avg	0.51	0.53	0.51	240

Table 1: ”CNN Baseline”

Emotion	Precision	Recall	F1-Score	Support
Neutral	0.45	0.88	0.60	16
Calm	0.61	0.69	0.65	32
Happy	0.52	0.34	0.42	32
Sad	0.53	0.25	0.34	32
Angry	0.64	0.84	0.73	32
Fearful	0.83	0.62	0.71	32
Disgust	0.68	0.66	0.67	32
Surprised	0.78	0.97	0.86	32
Accuracy			**0.64**	**240**
Macro Avg	0.63	0.66	0.62	240
Weighted Avg	0.64	0.64	0.62	240

Table 2: ”ResNet18”

The architectures were initially evaluated on the clean, uncorrupted test set to establish baseline predictive performance. The standard cross-entropy CNN baseline achieved an overall accuracy of 53% (Table 1). A review of its classification report reveals difficulties with lower-energy emotions, yielding F1-scores of 0.25 for ”Sad” and 0.23 for ”Happy”.

Emotion	Precision	Recall	F1-Score	Support
Neutral	0.47	0.44	0.45	16
Calm	0.67	0.69	0.68	32
Happy	0.33	0.22	0.26	32
Sad	0.28	0.31	0.29	32
Angry	0.68	0.72	0.70	32
Fearful	0.80	0.25	0.38	32
Disgust	0.58	0.66	0.62	32
Surprised	0.56	0.97	0.71	32
Accuracy			**0.54**	**240**
Macro Avg	0.55	0.53	0.51	240
Weighted Avg	0.55	0.54	0.52	240

Table 3: ”SupCon Frozen”

Emotion	Precision	Recall	F1-Score	Support
Neutral	0.35	0.69	0.47	16
Calm	0.66	0.72	0.69	32
Happy	0.39	0.38	0.38	32
Sad	0.64	0.22	0.33	32
Angry	0.69	0.75	0.72	32
Fearful	0.82	0.28	0.42	32
Disgust	0.68	0.84	0.75	32
Surprised	0.67	0.97	0.79	32
Accuracy			**0.60**	**240**
Macro Avg	0.61	0.61	0.57	240
Weighted Avg	0.63	0.60	0.57	240

Table 4: ”SupCon Fine-Tuned”

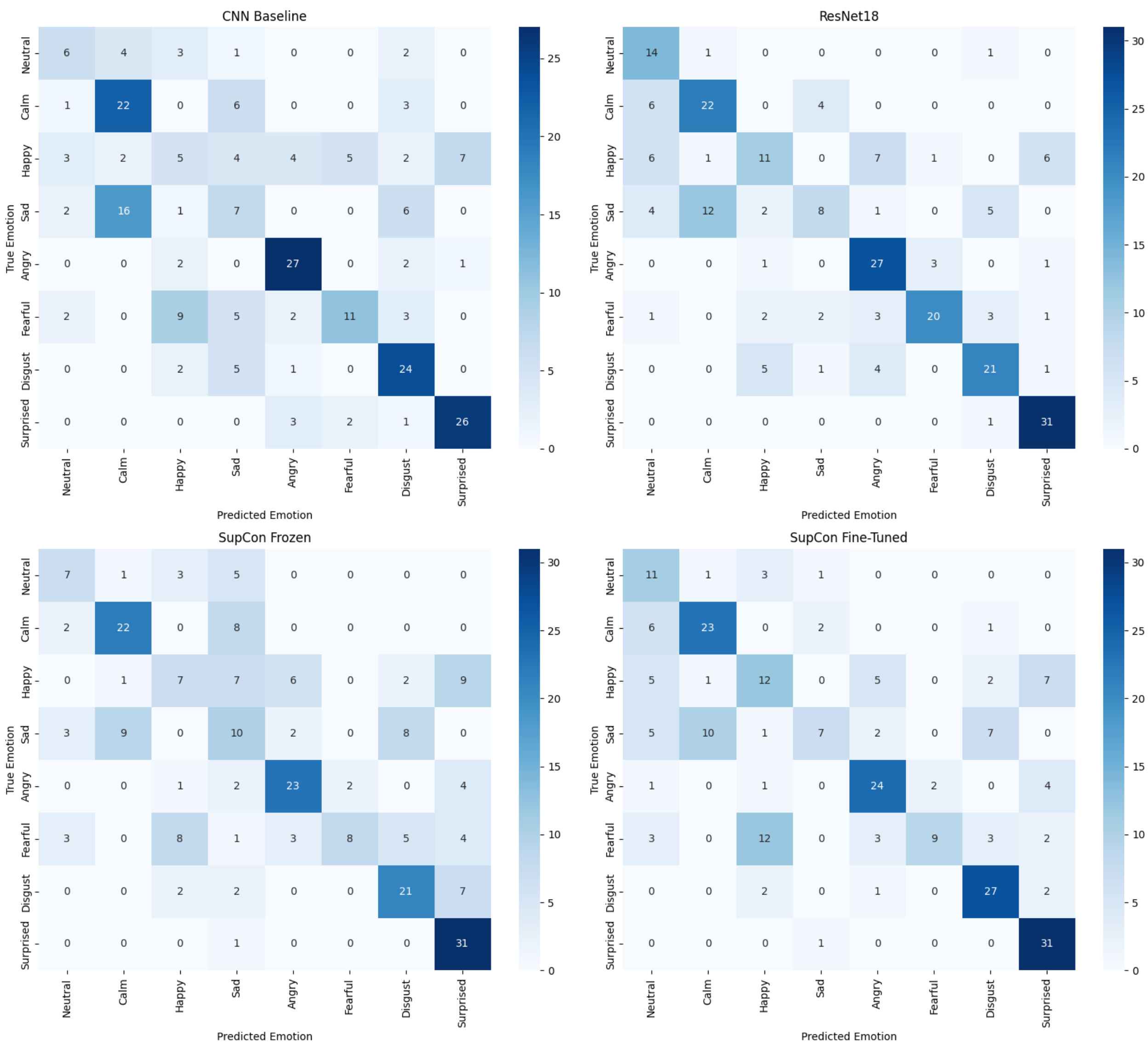

Figure 1: ”Side-by-side confusion matrices comparing the CNN Baseline and the Fine-Tuned SupCon model on the clean test set.”

Implementing the Supervised Contrastive Learning (SupCon) pipeline yielded structural benefits to the latent space. Evaluated via a linear probe, the frozen SupCon encoder achieved a 54% accuracy. While this represents only a marginal gain over the baseline, it proves that the Phase 1 contrastive pre-training successfully formed linearly separable clusters without any task-specific weight updates in the encoder.

Following end-to-end fine-tuning, the SupCon model reached a peak accuracy of 60% (Table 4), accompanied by an increase in the macro F1-score from 0.51 to 0.57. This 7% absolute accuracy improvement over the baseline suggests that contrastive clustering creates a more effective separation of emotional features, resulting in cleaner decision boundaries than point-wise Cross-Entropy optimization.

To contextualize the custom architecture's performance, a pre-trained ResNet18 was evaluated as a heavyweight benchmark, achieving the highest overall accuracy of 64% and a macro F1-score of 0.62. While this deep network predictably outperformed the lightweight custom networks in aggregate metrics, the SupCon Fine-Tuned model (60% accuracy) demonstrated highly competitive and sometimes superior class-specific clustering. Notably, the task-oriented contrastive pipeline allowed the much shallower custom encoder to outperform the massive ResNet18 architecture yielding higher F1-scores for "Calm" (0.66 versus 0.61) and "Sad" (0.64 versus 0.53).

4.2 Robustness Testing (The Gaussian Noise Stress Test)

To empirically validate the core hypothesis of this study the models were subjected to a Gaussian noise stress test. Increasing levels of static noise (Standard Deviation ranging from 0.0 to 0.5) were injected into the test set to simulate distributional shifts and signal degradation.

Noise STD	CNN Baseline (%)	ResNet18 (%)	SupCon Frozen (%)	SupCon Fine-Tuned (%)
0.0	53.33	64.17	53.75	60.00
0.05	50.42	65.42	53.33	56.25
0.1	49.58	65.83	45.00	47.92
0.2	25.83	62.50	32.08	29.17
0.3	17.50	58.33	28.75	23.33
0.4	13.75	43.33	17.08	26.67
0.5	13.33	35.00	13.75	19.58

Table 5: "Model robustness evaluation showing test accuracy degradation across increasing levels of Gaussian noise standard deviation."

The results (Table 5 and Figure 2) illustrate the performance limitations of the standard Cross-Entropy approach under noisy conditions. The CNN Baseline achieved an accuracy of 53.33% on clean audio but experienced a steep decline in predictive accuracy as noise increased. At 0.5 STD, performance degraded to 13.33%, a value that closely approximates random guessing across the eight distinct emotional classes (∼12.5%). This sharp degradation suggests that the baseline model may have overfitted to specific, localized frequency representations in the clean training data. Consequently, the introduction of Gaussian noise heavily disrupted its classification capability.

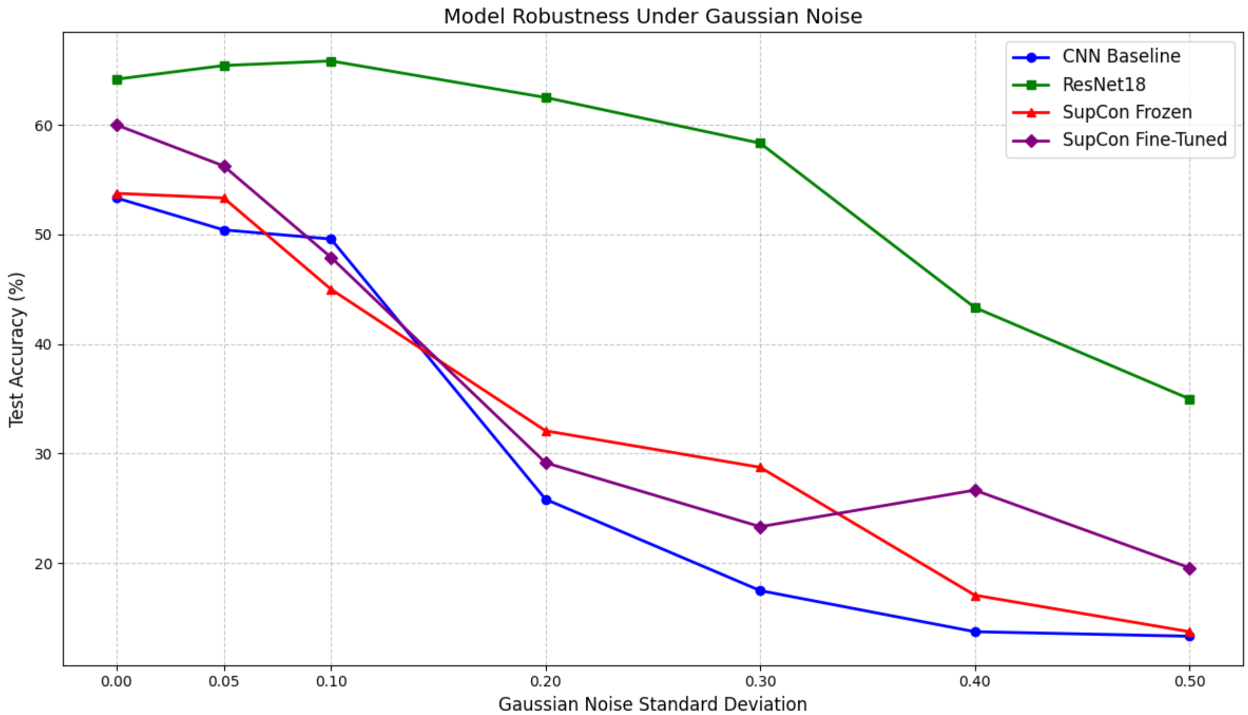

Figure 2: "Model robustness curve demonstrating the degradation of test accuracy across increasing Gaussian noise levels. The SupCon models exhibit a significantly slower rate of decay compared to the CNN Baseline."

In contrast, the architectures employing Supervised Contrastive Learning exhibited greater structural stability. The Fine-Tuned SupCon model established a higher baseline accuracy (60.00%) and showed a more gradual performance decline, maintaining 26.67% accuracy at 0.4 STD compared to the CNN Baseline's 13.75%. Additionally, the Frozen SupCon model recorded a higher accuracy at 0.3 STD (28.75%) than the Fine-Tuned variant (23.33%). This observation indicates that the pre-trained contrastive representations retain a higher degree of noise invariance. The subsequent end-to-end fine-tuning phase might have partially compromised this robustness by optimizing the linear classifier specifically for uncorrupted audio distributions.

These findings indicate that applying SpecAugment during pre-training helps the model learn general acoustic patterns rather than memorizing exact frequencies. Because the network depends less on isolated audio details, it groups similar emotions into clusters that remain distinct even when the signal is corrupted. This structural stability provides a measurable advantage over the baseline training approach.

5 Conclusion & Future work

This study demonstrates that Supervised Contrastive Learning (SupCon) provides a superior alternative to standard Cross-Entropy for Speech Emotion Recognition on limited datasets. By leveraging Multi-View masking via `SpecAugment`, the SupCon objective prevents the network from memorizing brittle, actor-specific acoustic frequencies. Instead, it forces the model to learn the generalized, invariant shape of human emotion. This approach not only elevated the baseline accuracy on clean audio, but also exhibited profound robustness against severe Gaussian noise, establishing contrastive pre-training as a highly effective methodology for deploying resilient audio processing models in noisy environments.

Future work will focus on three main areas to improve this pipeline. First, we will fine-tune the Mel-Spectrogram settings to extract ritch audio features. Second, we plan to combine ResNet18 with SupCon, adjusting the network so it processes audio frequencies effectively rather than treating them like standard images. Finally, because the RAVDESS dataset is small, we will expand the training data by generating synthetic samples using SMOTE and by enriching with the datasets from other languages.

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